

Cboe Europe
PITCH Specification

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1 Overview

This document describes the Cboe TCP PITCH protocol. The intended audience of this document is software developers of Cboe participants who would like to use the TCP PITCH protocol to receive real-time full depth of book quotations and execution information direct from Cboe, or those interested in receiving OTC trade reports or SI Quote data from the Cboe Trade Reporting Facility.

1.1 Trading Platforms

The TCP PITCH protocol used by Cboe on its European Trading platforms (BXE and CXE) is very similar to the PITCH protocol used in the US markets, but with the addition of new *long form* message types which can accomodate larger order sizes, wider symbols and finer grained prices. For bandwidth efficiency, long form messages are only sent if required. The European platforms also add a Trade - Extended message, which details off-book trades reported to Cboe, such as privately negotiated trades brought 'on-exchange'.

PITCH cannot be used to enter orders. For order entry, refer to the Cboe Europe FIX Specification.

All visible orders and executions are reflected via the PITCH feed. Each PITCH message reflects the addition, deletion, or execution of an order in the system. All orders and executions are anonymous, and do not contain the participant identity.

The following TCP PITCH streams are available:

Covered Markets	**Matching Units**
Full Feed	All
XLON, XDUB	1–3
XPAR, XAMS, XBRU, XLIS	4–6
XETR/XFRA, XWBO	7–8
MTAA/MTAH/ETFP, XMCE, CHIX[1], XWAR, XBUD, XPRA, XSWX, XVTX, XQMH	9–11
XCSE, XHEL, XSTO, XOSL	12

1.2 Trade Reporting Facility

The PITCH protcol is also used by the Cboe Trade Reporting Facility ("TRF") to disseminate OTC and Systematic Internaliser data.

Details of OTC or SI trades reported to the TRF system are published using the Trade - Extended message.

System Internaliser (SI) Quotes will be modelled in the TRF PITCH data using a variation of the existing PITCH Add Order messages. The Expanded Add Order message adds an attribution field allowing the quote to be attributed to a particular systematic internaliser, and a type field, which identifies the order as an SI Quote.

As a Systematic Internalizer modifies or cancels their existing quotes, this activity will be reflected on the PITCH feed as a series of Order Cancel and Expanded Add Order messages. Hence, participants who already have systems capable of processing Cboe PITCH messages may be able to re-use much of the same technology to maintain the current SI Quote book with minimal changes.

Order Executed and Trade messages are not applicable to the TRF PITCH feed.

[1] In addition all the primary MICs of secondary listed symbols.

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2 Protocol

2.1 Message Format

The PITCH data feed is comprised of a series of fixed-length sequenced messages. The length of each message is dependent upon the message type. Each message is made up of non-control ASCII bytes. Cboe reserves the right to add message types and grow the length of any message without notice. Participants should develop their decoders to ignore unknown message types and messages that grow beyond the expected length. Messages will only be grown to add additional data to the end of the message.

2.2 Sequence Numbers

The messages that make up the PITCH protocol are delivered using SOUP 2.0 which handles sequencing and delivery integrity. The Cboe implementation of SOUP 2.0 is the same as other market centres.

2.3 Order and Execution IDs

Note that Order and Execution IDs are 12 characters, base 36. The fourth character of an Execution ID may be used to differentiate type of execution:

Character	Meaning
0 (zero)	BXE Integrated Order Book
1	CXE Integrated Order Book
D	BXE Dark Order Book
E	CXE Dark Order Book

For example, the execution ID 000D3980018L is a BXE Dark Order Book trade.

Please note, use of the fourth character of an Execution ID to differentiate type of execution is now deprecated. Please utilise the Execution/Trade flags stamped on the message to achieve this differentiation.

2.4 Trade Amendments

Order-book or reported trades that are subsequently amended will result in two Trade - Extended Form messages to be sent. The first trade will be transmitted using all of the details of the original trade, including MMT flags, but with the Cancellation flag set. The second trade will be transmitted using the amended details, including MMT flags, but with the Modification flag set.

3 Sessions

The Cboe implementation of SOUP 2.0 is the same as other market centres. This includes the implementation of session events, such as login requests, client/server heartbeats, and logout requests.

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4 Message Types

Each PITCH message type reflects the addition, deletion, or execution of an order in the system.

The fields within each message type have a corresponding data type. The various data types are defined in the following table.

Data Type	Description
Alpha	A string of ASCII letters (A–Z), left justified and space padded on the right.
Alphanumeric	A string of ASCII numbers and letters (A–Z, 0–9), left justified and space padded on the right.
Base 36 Numeric	A string of ASCII numbers and letters (A–Z, 0–9), representing base 36 digits, right justified and zero filled on the left. Typically used for Order IDs and Execution IDs.
Numeric	A string of ASCII numbers (0–9), right justified and zero filled on the left.
Price	A string of ASCII numbers (0–9) consisting of six whole digits followed by four decimal digits. The whole number portion is zero filled on the left; the decimal portion is zero filled on the right. The decimal point is implied by position and does not explicitly appear in the field.
Long Price	As with Prices above, but this field consists of 12 whole number digits followed by seven decimal digits.
Timestamp	A string of ASCII numbers (0–9) representing the whole number of milliseconds past midnight London time, right justified and zero padded on the left, with no decimal point.

4.1 Modification of Orders

Messages which modify the state of an order (Order Executed and Order Executed — Long Form, Order Cancel and Order Cancel — Long Form) refer to an Order ID previously sent with an Add Order Message. Modifications are cumulative. Modification message reduce the remaining shares in the referenced open order by the number of shares indicated. When the remaining shares for an order reach zero, the order is dead and should be removed from the book.

4.2 System Event Message

Unlike other market centres, Cboe does not send System Event messages.

4.3 Symbol Clear

The `Symbol Clear` message instructs feed recipients to clear all orders for the Cboe book in the specified symbol. This message will be sent at startup each day. It would also be distributed in certain recovery events such as a data center fail-over.

The `Symbol Clear` message has the following format:

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"s"	Symbol Clear Message
Symbol	9	8	Alphanumeric	Symbol right padded with spaces.

4.4 Add Order Message

An add order message represents a newly accepted visible order on the Cboe book. It includes a day-specific Order ID assigned by Cboe to the order.

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Note: If an order is repriced within the Cboe matching engine, a Cancel Order Message will be immediately followed by an Add Order Message with the same Order ID as the original order.

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"A"	Add Order Message
OrderID	9	12	Base 36 Numeric	Day-specific identifier assigned to this order.
Side Indicator	21	1	Alpha	B = Buy S = Sell
Shares	22	6	Numeric	Number of shares being added to the book (may be less than the number of shares entered).
Symbol	28	6	Alphanumeric	Symbol, right added with spaces. Common Symbology Format.
Price	34	10	Price	Order price.
Display	44	1	"Y"	Always "Y". Maintained for compatibility reasons.

4.5 Add Order Message — Long Form

Only used on the Cboe European platform. As Add Order Message above, but with longer fields for Shares, Price and Symbol.

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"c"	Add Order Message — Long Form
OrderID	9	12	Base 36 Numeric	Day-specific identifier assigned to this order.
Side Indicator	21	1	Alpha	B = Buy S = Sell
Shares	22	10	Numeric	Number of shares being added to the book (may be less than the number of shares entered).
Symbol	32	8	Alphanumeric	Symbol, right added with spaces. Common Symbology Format.
Price	40	19	Long Price	Order price (long format).
Display	59	1	"Y"	Always "Y". Maintained for compatibility reasons.

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4.6 Expanded Add Order Message

The Expanded Add Order is used on the Cboe Europe TRF platform to provide visibility of Systematic Internalizer quotes. (SI Quotes are not executable via the TRF platform.)

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"t"	Add Order - Expanded
OrderID	9	12	Base 36 Numeric	Day-specific identifier assigned to this order
Side Indicator	21	1	Alpha	Valid values: B = Buy S = Sell
Quantity	22	10	Numeric	Number of shares applicable to this quote.
Symbol	32	8	Alphanumeric	Symbol, right padded with spaces
Price	40	19	Long Price	The quote price
Type	59	1	Alphanumeric	Attributed order type indicator. Currently defined values: S - 'SI' Quote
ParticipantID	60	4	Alphanumeric	Attributes this quote to a particular participant.

4.7 Order Executed Messages

Order Executed Messages are sent when a visible order on the Cboe book is executed in whole or in part. The execution price equals the price found in the original Add Order Message.

4.7.1 Order Executed Message

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"E"	Order Executed Message
OrderID	9	12	Base 36 Numeric	Order ID of a previously sent Add Order Message.
Executed Shares	21	6	Numeric	Number of shares executed.
Execution ID	27	12	Base 36 Numeric	Cboe generated day-unique execution identifier of this execution. Execution ID is also reference in the Trade Break Message.
Execution Flags	39	4	Alphanumeric	Type flags based on MMT v3.04 standard

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4.8 Order Executed Message — Long Form

Only used on the Cboe European platform. As Order Executed Message above, but with a longer field for Executed Shares.

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"e"	Order Executed Message — Long Form
OrderID	9	12	Base 36 Numeric	Order ID of a previously sent Add Order Message.
Executed Shares	21	10	Numeric	Number of shares executed.
Execution ID	31	12	Base 36 Numeric	Cboe generated day-unique execution identifier of this execution. Execution ID is also reference in the Trade Break Message.
Execution Flags	43	4	Alphanumeric	Type flags based on MMT v3.04 standard

4.8.1 Execution Flags

The Order Executed message uses a 4-character flags field to provide detailed type information regarding the execution.

Each character in the flags field corresponds to a distinct MMT field, as described in the following table and § 4.15, p. 17:

Execution Flags			
Offset	**MMT Level / Fieldname**		**Description**
0	1	Market Mechanism	See § 4.15, p. 17 for possible values
1	2	Trading Mode	
2	3.6	Ex/Cum Dividend	
3	3.9	Algorithmic Trade	

Implied MMT flags for the Order Executed message are as follows:

- *Level 1* populated per Execution Flags offset 0
- *Level 2* populated per Execution Flags offset 1
- *Level 3.1* will always be '-' for a standard trade
- *Level 3.2* will always be '-' for not being a Negotiated Trade
- *Level 3.3* will always be '-' for not being a Crossing Trade
- *Level 3.4* will always be '-' for no Modification Indicator
- *Level 3.5* will always be '-' for no Benchmark or Reference Price Indicator
- *Level 3.6* populated per Execution Flags offset 2
- *Level 3.7* will always be '-' for unspecified (as not off book)
- *Level 3.8* will always be 'P' for a Plain-Vanilla Trade
- *Level 3.9* populated per Execution Flags offset 3
- *Level 4.1* will always be '-' for no deferral of publication
- *Level 4.2* will always be '-' for not being applicable
- *Level 5* will always be '-' for not being applicable

4.9 Order Cancel Message

Order Cancel Messages are sent when a visible order on the Cboe book is cancelled in whole or in part.

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp

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Message Type	8	1	"X"	Order Cancel Message
OrderID	9	12	Base 36 Numeric	Order ID of a previously sent Add Order Message.
Cancelled Shares	21	6	Numeric	Number of shares cancelled.

4.10 Order Cancel Message — Long Form

Only used on the Cboe European platform. As Order Cancel Message above, but with a longer field for Cancelled Shares.

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"x"	Order Cancel Message — Long Form
OrderID	9	12	Base 36 Numeric	Order ID of a previously sent Add Order Message.
Cancelled Shares	21	10	Numeric	Number of shares cancelled.

4.11 Trade Message

The Trade Message provides information about executions of non-displayed orders or shares on the Cboe book and negotiated trades. Trade Messages are necessary to calculate Cboe execution based data. Trade Messages do not alter the book and can be ignored if you are just building a book.

No Add Order Message is sent for invisible orders or negotiated trades, and thus, no modify order messages may be sent when these orders are executed. Instead, a Trade Message is sent whenever such an order is executed in whole or in part. As with visible orders, hidden orders may be executed in parts.

A complete view of all Cboe executions can be built by combining all Order Executed and Trade messages.

The Order ID of a non-displayed order is obfuscated by default in the Trade Message but may be optionally disseminated for a Participant's own orders upon request. As such, partial executions against the same hidden order will by default have different Order IDs.

4.11.1 Trade Message

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"P"	Trade Message
OrderID	9	12	Base 36 Numeric	Obfuscated Order ID or Order ID of the invisible order or negotiated trade.
Side Indicator	21	1	Alpha	Always "B".
Shares	22	6	Numeric	Incremental number of shares executed.
Symbol	28	6	Alpha	Symbol, right padded with spaces.
Price	34	10	Price	Execution price of the fill.
Execution ID	44	12	Base 36 Numeric	Cboe generated day-unique execution identifier of this trade. Execution ID is also referenced in the Trade Break Message.
Trade Flags	56	5	Alphanumeric	Type flags based on MMT v3.04 standard

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4.11.2 Trade Message — Long Form

Only used on the Cboe European platform. As Trade Message above, but with longer fields for Shares and Price.

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"q"	Trade Message — Long Form
OrderID	9	12	Base 36 Numeric	Obfuscated Order ID or Order ID of the invisible order or negotiated trade.
Side Indicator	21	1	Alpha	Always "B".
Shares	22	10	Numeric	Incremental number of shares executed.
Symbol	32	8	Alpha	Symbol, right padded with spaces.
Price	40	19	Long Price	Execution price (long format) of the fill.
Execution ID	59	12	Base 36 Numeric	Cboe generated day-unique execution identifier of this trade. Execution ID is also referenced in the Trade Break Message.
Trade Flags	71	5	Alphanumeric	Type flags based on MMT v3.04 standard

4.11.3 Trade Flags

The non-Extended Trade messages use a 5-character flags field to provide detailed type information regarding the trade.

Each character in the flags field corresponds to a distinct MMT field, as described in the following table and § 4.15, p. 17:

Trade Flags			
Offset	**MMT Level / Fieldname**		**Description**
0	1	Market Mechanism	See § 4.15, p. 17 for possible values
1	2	Trading Mode	
2	3.1	Transaction Category	
3	3.5	Benchmark/Reference Price Indicator	
4	3.9	Algorithmic Trade	

Implied MMT flags for the non-Extended Trade messages are as follows:

- *Level 1* populated per Trade Flags offset 0
- *Level 2* populated per Trade Flags offset 1
- *Level 3.1* populated per Trade Flags offset 2
- *Level 3.2* will always be '-' for not being a Negotiated Trade
- *Level 3.3* will always be '-' for not being a Crossing Trade
- *Level 3.4* will always be '-' for no Modification Indicator
- *Level 3.5* populated per Trade Flags offset 3
- *Level 3.6* will always be '-' for no Special Dividend
- *Level 3.7* will always be '-' for unspecified (as not off book)
- *Level 3.8* will always be 'P' for a Plain-Vanilla Trade
- *Level 3.9* populated per Execution Flags offset 4
- *Level 4.1* will always be '-' for no deferral of publication
- *Level 4.2* will always be '-' for not being applicable
- *Level 5* will always be '-' for not being applicable

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4.11.4 Trade Message — Extended Form

Only used on the Cboe European platform. This provides extended details of trades reported to or executed by Cboe. This includes, for example, privately negotiated trades brought 'on-exchange'. Like other Trade messages, these do not alter the book, and can be ignored if you are just building a book.

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"O"	Trade Message — Extended Form
Shares	9	12	Numeric	Number of shares executed.
Symbol	21	8	Alphanumeric	Symbol, right padded with spaces.
Price	29	19	Long Price	Execution price of the fill. This may be zero if the price is pending, as denoted by Level 3.8 of the Extended Trade Flags.
Trade ID	48	12	Base 36 Numeric	Cboe generated identifier of this trade. This identifier is guaranteed to be unique for at least 7 calendar days.
Trade Date	60	8	Alphanumeric	Date on which the trade occurred, in the form YYYYMMDD.
Trade Time	68	8	Timestamp	Time at which the trade occurred, encoded as the number of milliseconds since midnight, on the day represented by `Trade date`, London time.
Execution Venue	76	4	Alphanumeric	The venue on which the trade executed, when applicable. This will contain the MIC representing the venue on which the trade occurred, where applicable. e.g. for Cboe NT trades, this shall be the segment MIC BATF for BXE off-book trades, CHIO for CXE off-book trades and BARO for REGM off-book trades. This will contain SINT if the trade occurred on a Systematic Internaliser or XOFF if OTC. Cboe LIS trades have the value LISX.
Currency	80	3	Alphanumeric	Traded currency.
Cboe Trade Timing Indicator	83	1	Alphanumeric	'1' The trade was reported to Cboe 'late', '2' The trade was reported to Cboe out of the Main Session, '3' The trade was reported to Cboe late and out of the Main Session, '-' otherwise.
Extended Trade Flags	84	14	Alphanumeric	Type flags based on the MMT v3.04 standard.

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4.11.5 Trade Message — Unknown Symbol

Only used on the Cboe European Trade Reporting Facility. This provides details of trades reported to Cboe, but traded on a symbol not currently known to Cboe. These trades are identified by the ISIN and the reported currency. Like other Trade messages, these do not alter the book, and can be ignored if you are just building a book.

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"k"	Trade Message — Unknown Symbol
Shares	9	12	Numeric	Number of shares traded.
Symbol	21	12	Alphanumeric	Symbol in ISIN
Price	33	19	Long Price	Reported price of the trade. This may be zero if the price is pending, as denoted by Level 3.8 of the Extended Trade Flags.
Trade ID	52	12	Base 36 Numeric	Cboe generated identifier of this trade. This identifier is guaranteed to be unique for at least 7 calendar days.
Trade Date	64	8	Alphanumeric	Date on which the trade occurred, in the form YYYYMMDD.
Trade Time	72	8	Timestamp	Time at which the trade occurred, encoded as the number of milliseconds since midnight, on the day represented by `Trade date`, London time.
Execution Venue	80	4	Alphanumeric	The venue on which the trade executed, when applicable. This will contain the MIC representing the venue on which the trade occurred, where applicable. The field will contain SINT if the trade occurred on a Systematic Internaliser or XOFF if OTC.
Currency	84	3	Alphanumeric	Reported currency.
Cboe Trade Timing Indicator	87	1	Alphanumeric	'1' The trade was reported to Cboe 'late', '2' The trade was reported to Cboe out of the Main Session, '3' The trade was reported to Cboe late and out of the Main Session, '-' otherwise.
Extended Trade Flags	88	14	Alphanumeric	Type flags based on the MMT v3.04 standard.

4.11.6 Extended Trade Flags

The Cboe Extended Trade message uses a 14character flags field to provide detailed type information regarding the trade.

Each character in the flags field corresponds to a distinct MMT field, as described in the following table and § 4.15, p. 17:

Trade Type Flags			
Offset	**MMT Level / Fieldname**		**Description**
0	1	Market Mechanism	See § 4.15, p. 17 for possible values.
1	2	Trading Mode	
2	3.1	Transaction Category	
3	3.2	Negotiated Trade	
4	3.3	Crossing Trade	
5	3.4	Modification Indicator	
6	3.5	Benchmark/Reference Price Indicator	
7	3.6	Special Dividend	
8	3.7	Off Book Automated Indicator	
9	3.8	Price Formation/Discovery Process	
10	3.9	Algorithmic Indicator	
11	4.1	Publication Mode/Deferral Reason	
12	4.2	Deferral or Enrichement Type	
13	5	Duplicative Indicator	

Special notes regarding Deferral or Enrichment Type

This is for RTS 2 only and currently unsupported in Cboe. A value of "-" should hence be expected for offset 12 (level 4.2).

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4.12 Trading Status Message

The Trading Status Message is used to indicate the current trading status of a security. A Trading Status Message will be sent whenever a security's trading status changes. In addition, Cboe will send a Trading Status Message for all securities that are "Suspended" before the start of trading hours.

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"H"	Trading Status Message
Symbol	9	8	Alphanumeric	Symbol right padded with spaces.
Status	17	1	Alpha	"T" = Trading "R" = Off-Book Reporting "C" = Closed "S" = Suspended "N" = No Reference Price "V" = Volatility Interruption "O" = Opening Auction "E" = Closing Auction "H" = Halt[2] "M" = Market Order Imbalance "P" = Price Monitoring Extension
Reserved	18	3	Alpha	Reserved

See the Participant Manual for details on Trading Status phases.

4.13 Statistics Message

Only used on the Cboe European platform. The Statistics Message is used to disseminate the statistics prices: opening, closing, high, low. A Statistics Message will be sent whenever a value changes. At the start of each trading day a "Previous Closing Price" will be sent with the closing price of the previous trading day.

If a trade that generated the price is subsequently busted another Statistics Message will be sent.

The "Price Determination" will by default be "Normal". The value of "Manual" arises from prices being adjusted by market supervision. A lower "High Price" or higher "Low Price" could result from breaking a trade, these will be flagged with "Manual".

Cboe reserves the right to add additional values to the "Statistics Type" and "Price Determination" fields without notice. Participants should develop their decoders to ignore unknown values.

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"Z"	Statistics Message
Symbol	9	8	Alphanumeric	Symbol right padded with spaces.
Price	17	19	Long Price	Price.
Statistic Type	36	1	Alpha	"C" = Closing Price "H" = High Price "L" = Low Price "O" = Opening Price "P" = Previous Closing Price

[2]Reserved for future use

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Price Determination	37	1	Alphanumeric	"0" = Normal "1" = Manual (Price override by Market Supervision)

4.14 Auction Messages

4.14.1 Auction Update Message

Auction Update messages are used to disseminate indicative price and size information during auctions for Cboe auction eligible securities. For Opening and Closing auctions, Auction Update messages are published periodically during the call and extension phases of the auction process. For Periodic auctions, they are published when there is material change.

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"I"	Auction Update Message
Symbol	9	8	Alphanumeric	Symbol right padded with spaces.
Auction Type	17	1	Alpha	"O" = Opening Auction "C" = Closing Auction "H" = Halt Auction[3] "V" = Volatility Auction[3] "P" = Periodic Auction
Reference Price	18	19	Long Price	Reference price used in tie-breaker situations
Indicative Price	37	19	Long Price	Price at which the auction would match if executed at the time of the message
Indicative Shares	56	10	Numeric	Number of shares at the Indicative Price
Outside Tolerance	66	1	Alphanumeric	Indicates whether the price on this update is outside the Cboe EBBO collar: "O" = Outside tolerance "I" = Inside tolerance "-" = Not specified
Includes Primary	67	1	Alphanumeric	Indicates whether the Cboe EBBO used to collar this update includes the Primary Market quotes: "P" = Includes Primary "N" = Excludes Primary "-" = Not specified

4.14.2 Auction Summary

Auction Summary messages are used to disseminate the results of an auction in a Cboe auction eligible security.

Field Name	Offset	Length	Value	Description
Timestamp	0	8	Timestamp	Timestamp
Message Type	8	1	"j"	Auction Summary Message
Symbol	9	8	Alphanumeric	Symbol right padded with spaces.

[3]Reserved for future use

Auction Type	17	1	Alpha	"O" = Opening Auction "C" = Closing Auction "H" = Halt Auction[4] "V" = Volatility Auction[4] "P" = Periodic Auction
Price	18	19	Long Price	Auction price
Shares	37	10	Numeric	Cumulative number of shares executed during the auction

[4]Reserved for future use

4.15 MMT Value Mappings

The following tables define type information as detailed by version 3.04 of the Market Model Typology standard ("MMT"). See `http://www.fixtradingcommunity.org/pg/group-types/mmt` for more details.

Not all values are currently applicable to Cboe services. However, participants are advised to design their systems to cope with any of the listed MMT values.

1. Market Mechanism	
Value	**Meaning**
'1'	Central Limit Order Book
'2'	Quote Driven Market
'3'	Dark Order Book
'4'	Off Book
'5'	Periodic Auction
'6'	Request For Quotes
'7'	Any Other, Including Hybrid

2. Trading Mode	
Value	**Meaning**
'1'	Undefined Auction
'2'	Continuous Trading
'3'	At Market Close Trading
'4'	Out Of Main Session
'5'	Trade Reporting (On Exchange)
'6'	Trade Reporting (Off Exchange)
'7'	Trade Reporting (Systematic Internalizer)
'O'	Scheduled Opening Auction
'K'	Scheduled Closing Auction
'I'	Scheduled Intraday Auction
'U'	Unscheduled Auction

3.1 Transaction Category	
Value	**Meaning**
'D'	Dark Trade
'R'	Trade that has Received Price Improvement
'Z'	Packaged trade
'Y'	Exchange for Physicals Trade
'-'	None of the above apply

3.2 Negotiated Trade or Pre-Trade Transparency Waiver	
Value	**Meaning**
'1'	Negotiated Trade in Liquid Financial Instruments
'2'	Negotiated Trade in Illiquid Financial Instruments
'3'	Negotiated Trade Subject to Conditions Other than the Current Market Price
'N'	Negotiated Trade Where None of the Above Apply
'4'	Pre-Trade Transparency Waiver for Illiquid Instrument on an SI
'5'	Pre-Trade Transparency Waiver for Above Standard Market Size on an SI
'6'	Pre-Trade Transparency Waivers for Illiquid Instrument on an SI and Above Standard Market Size on an SI
'-'	Not specified

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3.3 Crossing Trade	
Value	**Meaning**
'X'	Crossing Trade
'-'	Not specified

3.4 Modification Indicator	
Value	**Meaning**
'A'	Indicates a modification of a previously reported trade
'C'	Indicates a cancellation of a previously reported trade
'-'	Not specified

3.5 Benchmark or Reference Price Indicator	
Value	**Meaning**
'B'	Benchmark trade if (optionally) set by reporting party
'S'	Reference Price Trade
'-'	Not specified

3.6 Ex/Cum Dividend	
Value	**Meaning**
'E'	Ex/Cum/Special dividend if (optionally) set by reporting party
'-'	Not specified

3.7 Off Book Automated Indicator	
Value	**Meaning**
'Q'	Automated
'M'	Manual
'-'	Not specified

3.8 Contribution to Price Formation or the Price Discovery Process	
Value	**Meaning**
'P'	Standard trade for the specified Market Mechanism or Trading Mode
'T'	Non-Price Forming Trade (formerly known as Technical Trade)
'J'	Trade not Contributing to Price Discovery Process (formerly Technical Trade)
'N'	Price is currently not available but pending

3.9 Algorithmic Trade	
Value	**Meaning**
'H'	Algorithmic Trade
'-'	Non-algorithmic Trade

4.1 Publication Mode / Post-Trade Deferral Reason	
Value	**Meaning**
'1'	Trade report reported late without permitted deferral
'2'	Deferral Trade for "Large In Scale"
'3'	Deferral Trade for "Illiquid Instrument"
'4'	Deferral Trade for "Size Specific"
'5'	Deferral Trade for "Illiquid Instrument" and "Size Specific"
'6'	Deferral Trade for "Illiquid Instrument" and "Large In Scale"
'-'	Not specified (Immediate Publication)

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4.2 Post-Trade Deferral or Enrichment Type	
Value	**Meaning**
'1'	Limited Details Trade
'2'	Daily Aggregated Trade
'3'	Volume Omission Trade
'4'	Four Weeks Aggregation Trade
'5'	Indefinite Aggregation Trade
'6'	Volume Omission Trade, Eligible For Subsequent Enrichment in Aggregated Form
'7'	Full Details of Earlier Limited Details Trade
'8'	Full Details of Earlier Daily Aggregated Trade
'9'	Full Details of Earlier Volume Omission Trade
'V'	Full Details of Four Weeks Aggregation Trade
'W'	Full Details of Earlier Volume Omission Trade, Eligible For Subsequent Enrichment in Aggregated Form
'-'	Not Applicable

5. Duplicative Indicator	
Value	**Meaning**
'1'	Duplicative Trade Report
'-'	Unique Trade Report

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5 Support

Please email support questions or comments regarding this specification to `tradedeskeurope@cboe.com`.

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Revision History

11 July 2008	Initial draft version.
16 July 2008	Updated contact information.
12 November 2008	Changed data type for symbol from Alpha to Alphanumeric. Updated front page text to note FSA authorisation.
2 December 2008	Noted that Trade Break Messages are not currently sent.
31 July 2009	Trade Message and Trade Message — Long Form will always have a Side Indicator of "B".
15 December 2009	Version 2.0. Added Order and Execution IDs subsection which notes how to infer different types of executions (e.g., Cboe Dark Pool).
17 March 2010	Version 2.1 Corrected message type of Cancel Order — Long Form message.
20 April 2010	Version 2.2 Clarified size of prices on long form messages by noting that their value is "Long Price" as opposed to "Numeric".
20 May 2010	Version 2.3 Trade Message and Trade Message — Long Form will always have an Order ID of "000000000000" for dark book orders.
18 June 2010	Version 2.4 Order IDs in Trade Messages are now obfuscated by default. This obsoletes the change made to Order IDs on 20 May 2010.
16 September 2010	Version 2.5 Noted that two TCP PITCH streams (all instruments or just unit 12 instruments) are available in the Overview.
5 October 2010	Version 2.6 Documentation incorrectly stated that dark book orders will have an OrderID consisting of all zeroes. In actuality, dark book OrderIDs are obfuscated just like any other hidden order. No functional changes.
11 July 2011	Version 2.7 Noted that trade break messages *are* sent.
21 September 2011	Version 2.8 Removed references to FAST.
13 December 2011	Version 2.9 Removed references to MOC/TAL.
12 January 2012	Version 2.10 Updated split feeds to reflect newly added streams.
22 February 2012	Version 2.11 Added XFRA and ETFP MICs to Overview.
4 April 2012	Version 2.12 Moved WBAH to the same stream as XETR/XFRA, units 8-9. Updated to reflect the availability of units 9-11 as a single stream (available from April 16th 2012).
17 April 2012	Version 2.13 Added Chi-X Europe Integrated and Dark Pool Execution ids.
7 February 2013	Version 3.0 New Off-Book Trade, Off-Book Trade Break and Symbol Clear messages.
21 March 2013	Version 3.1 Off-Book Trade size will be 8 digits, Off-Book Trade price field will be 19 digits. Support for indicating an off-book trade was reported out of the Main Session.

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9 April 2013	Version 3.2 Updated link to FESE website
20 June 2013	Version 3.3 The introduction has been re-worded, and section 1.2 added, introducing use of PITCH by the Trade Reporting Facility. Addition of Expanded Add Order message to support SI Quote publication. Addition of Trading Status and Statistics messages.
1 August 2013	Version 3.4 Addition of CHIX to MIC codes in the symbol ranges.
15 August 2013	Version 3.5 Additional information given on the new Trading Status and Statistics messages.
19 September 2013	Version 3.6 MIC update: WBAH to XWBO
6 December 2013	Version 4.0 Renamed Off-Book Trade Message to Trade Report Message. Removed the Off-Book Trade Break Message. Use a Modification Indicator of 'C' in the Trade Type Flags field to delete a Trade Report. Widened the Shares field from 8 to 12 bytes in the Trade Report (previously Off-Book Trade) message. Added a special value of 'BCS' to the Execution Venue field in the Trade Report (previously Off-Book Trade) message to indicate a 'Broker Crossing System' trade. Added a new Cboe specific Transaction Sub-Category flag to the end of the Trade Report Flags field in the Trade Report (previously Off-Book Trade) message. Widened the Symbol field from 6 to 8 bytes in the Add Order - Long and Trade - Long messages. Expanded the Status flag in the Trading Status message to include values for Regulatory Halts ('H'), Market Order Imbalance ('M') and Price Monitoring ('P') extensions. Additionally the Auction ('A') status has been removed and sub-divided into Opening Auction ('O') and Closing Auction ('E'). Added an Execution Flags field to the Order Executed and Order Executed – Long messages. Added a Trade Flags field to the Trade message. Section 4.14 added, introducing the Auction Update and Auction Summary messages.
24 December 2013	Version 4.1 Made Trade Report Message's Transaction Sub-Category field reserved for future use.
21 January 2014	Version 4.2 Renamed 'Regulatory Halt' trading status to 'Halt'. Clarified the trading statuses that are reserved for future use. Corrected the implied value for the level 3.1 MMT flag in the Execution Flags section. Changed the shares field width to 10 bytes in both the Auction Update and0 Auction Summary messages to bring them in line with the Multicast PITCH specification. Changed the Auction Update message type from 'i' to 'k'. Changed the Trade — Long Form message type from 'p' to 'q'. Changed the Add Order — Long Form message type from 'a' to 'c'. Changed VenueField value of BCS to AUT.

12 June 2014	Version 4.3 Deprecate usage of the fourth character of the Execution ID to help differentiate the nature of the trade in favour of MMT flags directly. Rename the Cboe Transaction Sub-Category Trade Report flag for the new MMT 3.7 trade flag. Added support for the new MMT 2 flag for an undefined auction.
23 September 2014	Version 4.3.1 Add XIST (Turkey) to the units 9–11 stream.
23 September 2014	Version 4.4 Removed 'effective from' labels.
4 February 2015	Version 4.4.1 Corrected timestamp data type from Numeric to Timestamp.
29 March 2015	Version 4.4.2 Clarify trade timing indicator.
8 June 2015	Version 4.5 Remove deprecated AUT flag. Rename Trade Report to Trade - Extended. New Auction Update message type. Extended the Auction Type flag in the Auction Update and Auction Summary messages to include Periodic Auctions ('P').
14 December 2015	Version 4.5.1 Add XQMH to the units 9–11 stream.
14 December 2015	Version 4.6 SINT and XOFF as possible values for the Execution Venue field of the Trade - Extended Form message. Details on semantic change for Trade Amendments. Removal of Trade Break messages.
19 February 2016	Version 4.7 Updated for new branding.
29 April 2016	Version 4.8 Removed 'Effective' content related to Q2 2016 Release.
17 June 2016	Version 4.9 Renamed a few dangling 'Trade Reporting message' to 'Trade - Extended message'.
8 July 2016	Version 4.10 Added MTAH market
1 February 2017	Version 4.11 MMT v3 support
23 May 2017	Version 4.12 Clarified MIC used in Execution Venue for Cboe LIS trades.
19 July 2017	Version 4.13 MMT v3.04 support for Q4 2017 release.
2 August 2017	Version 4.14 Removed the XIST (Turkey) as a supported market.
26 January 2018	Version 4.15 Updated MIC used in Execution Venue for Cboe NT trades in BXE and CXE.
31 August 2018	Version 4.16 Add new markets (XWAR, XBUD & XPRA) to unit 12.

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